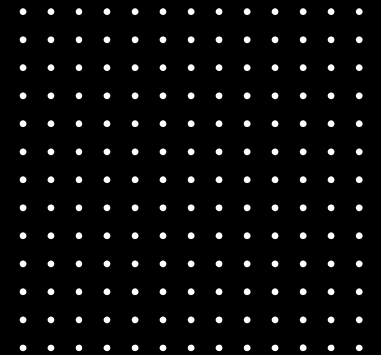




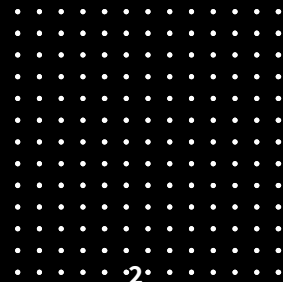
Detecting upi fraud Transactions



Introduction



- The rapid rise of UPI transactions has increased fraud risks.
- Machine learning helps detect fraudulent transactions in real time.
- This study applies Random Forest to classify transactions as fraudulent or legitimate.
- Evaluate the model's effectiveness using accuracy, precision, recall, and F1-score.



Methodology



Data Preprocessing:

Loaded and cleaned UPI fraud dataset.

Handled missing values.

Extracted time-based features from transaction timestamps.

Removed non-numeric and unnecessary features.

Model Training:

Baseline Model: Trained Random Forest on imbalanced data.

Improved Model: Applied SMOTE to balance fraudulent and non-fraudulent transactions.

Split data into 80% training and 20% testing.

Evaluation Metrics:

Accuracy, Precision, Recall, F1-score.

Confusion Matrix for performance analysis.



Hyperparameters



Model Used: Random Forest Classifier

- `n_estimators = 100` → Number of decision trees in the forest.
- `max_depth = None` → Trees grow until all leaves are pure (or a stopping condition is met)
- `min_samples_split = 2` → Minimum samples required to split a node.
- `min_samples_leaf = 1` → Minimum samples required at a leaf node.
- `random_state = 42` → Ensures reproducibility.
- `bootstrap = True` → Uses bootstrapped samples for training each tree
- `sampling_strategy = 'auto'` (SMOTE) → Balances the dataset by generating synthetic fraud cases.



RESULT

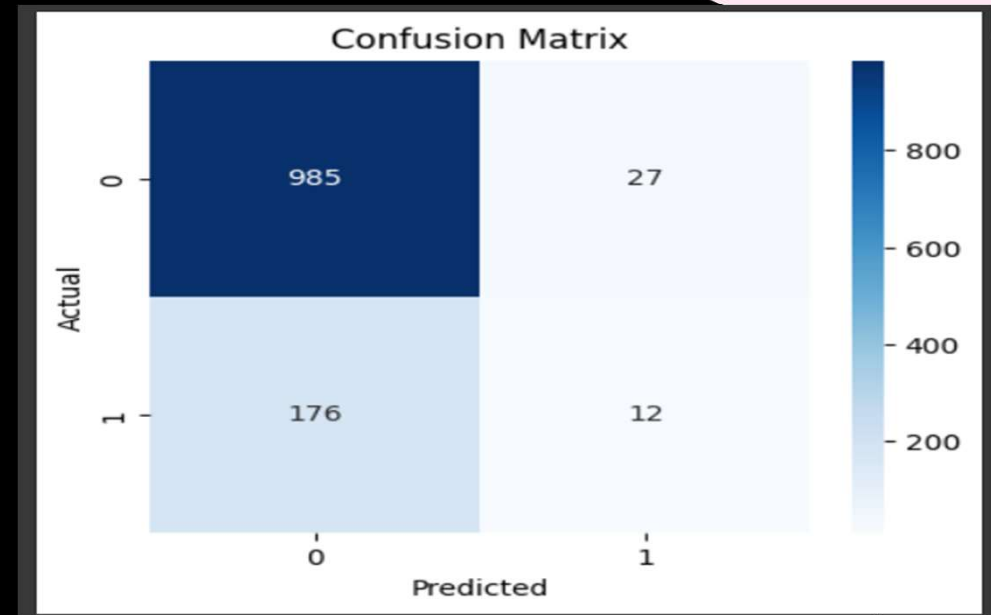
Baseline Model (Before SMOTE):

Accuracy: higher accuracy but imbalanced data.

Confusion matrix showed poor fraud detection.



```
✓ [26] accuracy_score(y_test, y_pred)  
→ 0.8308333333333333
```



After Applying SMOTE:

- Accuracy improved.
- Better fraud detection rates.
- Improved recall and F1-score for the fraudulent class.



Model Accuracy: 78.76%

Classification Report (in %):

Class 0:

Precision: 80.59%

Recall: 75.79%

F1-Score: 78.12%

Support: 100800

Class 1:

Precision: 77.13%

Recall: 81.73%

F1-Score: 79.36%

Support: 100700

Class macro avg:

Precision: 78.86%

Recall: 78.76%

F1-Score: 78.74%

Support: 201500

Class weighted avg:

Precision: 78.86%

Recall: 78.76%

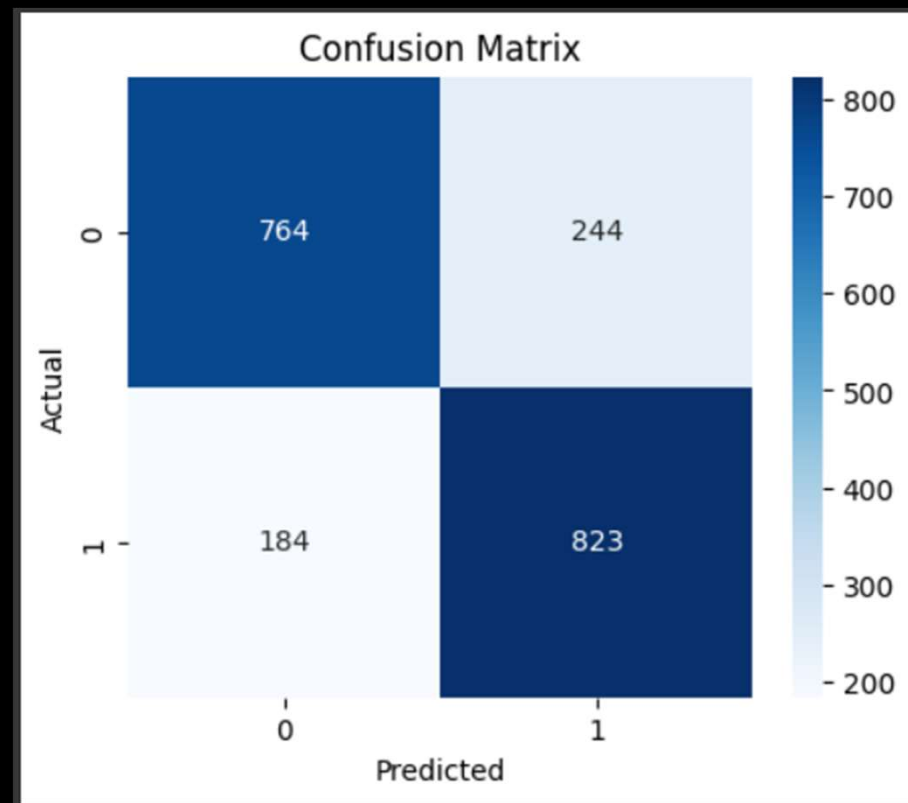
F1-Score: 78.74%

Support: 201500

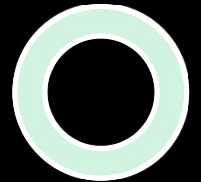


Confusion Matrix Comparison:

More balanced true positives and true negatives.



Final result



```
✓ 0s # Ensure the new transaction has all required features
# Get the list of columns from the training data (X)
columns = X.columns.tolist()

# Create the new transaction DataFrame with matching columns and data
new_transaction = pd.DataFrame([[
    14150, # Example transaction amount
    4,    # time of transaction
    0,    # Day of transaction
    26,   # Example: Transaction Date
    1,    # Example: Month
    2023, # Example: Year
    29800, # zip code
    0,    # Example value for an additional column (replace with actual value)
    0,    # Example value for an additional column (replace with actual value)
    0     # Example value for an additional column (replace with actual value)
]], columns=columns) # Ensure column names match training data

# Step 3: Standardize the new transaction using the same scaler
# Before standardizing, make sure the scaler is fitted using the training data (X)
# Example: scaler = StandardScaler().fit(X)
# new_transaction = scaler.transform(new_transaction)

# Step 4: Predict fraud
fraud_prediction = model.predict(new_transaction)

# Step 5: Print the result
print("Fraudulent Transaction" if fraud_prediction[0] == 1 else "Legitimate Transaction")
```

Fraudulent Transaction

The model accurately detects fraudulent UPI transactions using machine learning. This enhances financial security by enabling real-time fraud prevention

Inferences



- Applying SMOTE significantly improved fraud detection.
- The Random Forest model performed better on balanced data.
- Handling class imbalance is crucial for effective fraud detection.
- Further improvements could involve feature engineering and ensemble models.





Thank you